

agent: discovery-code-quality-agent llm: claude-opus-4-6 run_id: 20260930T121134_g4zvb3 generated_at: 2026-09-30T06:58:03Z

2. Code Quality & Complexity Hotspots Analysis

Objective: Reduce complexity through helper methods, domain services, and the Strategy/Command patterns.

Date: 2026-09-30 06:58:03 UTC | **Scope:** shende-shweta/pingcrm (master) — Laravel 12 / PHP 8.3 (backend) + React / TypeScript / Inertia.js (frontend) + MySQL

Executive Summary

EXECUTIVE SUMMARY

The pingcrm codebase comprises 319 source files totalling ~181,000 LOC across a Laravel 12 backend (141 PHP files, 77,262 LOC) and a React/TypeScript frontend (137 TS/TSX files, 100,446 LOC). An estimated 84% of the codebase consists of structurally duplicated IVR (Interactive Voice Response) module code: 81 near-identical 759-line PHP controllers, 12 "GodService" classes, 133 templated frontend page files, 229 legacy monolith components, and 8 duplicated utility modules. The original Pingcrm application (contacts, organizations, users, dashboard) is cleanly structured, but the IVR layer introduces severe code-quality risks including SQL injection via string concatenation in every IVR controller, 4,400 uses of PHP's `extract()` on unvalidated request data, 12 hardcoded API keys, and mutable static state that would leak memory under long-running process models. Git churn is low (118 total commits) and ownership is concentrated, so defect density and coordination risks are currently minimal — but the extreme duplication means that any bug fix or security patch must be replicated across 80+ files manually.

319

FILES ANALYZED

0

FUNCTIONS/METHODS OVER 200 LOC

89

CLASSES/FILES OVER 300 LOC

~12

HIGHEST CYCLOMATIC COMPLEXITY

OVERALL CODEBASE RATING — CODE QUALITY & COMPLEXITY

High Risk

Driven by catastrophic duplication (H4/H5 both >80%) and large generated files exceeding 1,000 LOC (H2), despite healthy churn and ownership metrics.

HOTSPOT SCORE (WEIGHTED COMPOSITE)

39 / 100 — Moderate

Hotspot Score = (Cyclomatic Complexity 50 × 25%) + (Code Churn 15 × 25%) + (Defect Density 15 × 20%) + (Class/Function Size 68 × 15%) + (Business Logic Duplication 95 × 10%) + (Developer Ownership Risk 10 × 5%) =

$$12.5 + 3.75 + 3.0 + 10.2 + 9.5 + 0.5 = 39$$

The weighted Hotspot Score (39) falls in the Moderate band because healthy churn, defect-density, and ownership metrics (50% of weight) dilute the extreme duplication. However, three hotspots independently rate High Risk (H2, H4, H5), so the worst-wins Overall Rating is **High Risk**.

2.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	~12 (LoadslvrModuleData trait, match + query builder chains)	MODERATE
H2	Large Classes	Largest class/file LOC	<300	300–1000	>1000	1,101 LOC (legacyFormatters*.ts); 759 LOC (81 IVR controllers)	HIGH RISK
H3	Large Functions	Largest function/method LOC	<50	50–200	>200	~50 LOC (loadCallRows in LoadslvrModuleData)	GOOD
H4	Business Logic Duplication	Duplicated business logic %	<5%	5–10%	>10%	~84% — IVR controller/service/repo/model chain duplicated 12x across domains	HIGH RISK
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	~84% — 152,798 of 181,029 LOC are near-identical copies	HIGH RISK
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	~2 (README.md most changed; 118 total commits)	GOOD
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	2 (photo upload fixes are the densest cluster)	GOOD
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	>95% (IVR layer single-author; original Pingcrm: 3 primary authors)	GOOD
H9 (additional)	SQL Injection	Controllers with raw string-concat SQL	0	1–5	>5	83 IVR controllers with DB::select string concatenation	HIGH RISK
H10 (additional)	Unsafe extract()	Total extract() call sites	0	1–10	>10	4,400 occurrences across IVR controllers and GodServices	HIGH RISK
H11 (additional)	Static Mutable State	GodService classes with static cache	0	1–3	>3	12 GodServices each with public static \$sharedRuntimeCache	HIGH RISK

H12 (additional)	Hardcoded Secrets	Files with hardcoded API keys	0	1–2	>2	12 GodServices each with hardcoded \$apiKey strings	HI
C1 (context)	Unmanaged Static State	Static arrays accumulating data	0	1–3	>3	12 GodServices — \$sharedRuntimeCache never cleared; would leak under Octane/Swoole	HI
C2 (context)	Unbounded Queries	Controllers using ->get() without limits	0	1–5	>5	83 IVR controllers use ->get() on full tables; no chunking or pagination	HI
C3 (context)	Blocking sleep()	Services with synchronous sleep()	0	1–5	>5	All 12 GodServices call sleep(1) in every method (~300 call sites)	HI
C4 (context)	Laravel Octane / Long-Running	n/a	n/a	n/a	n/a	Requested via context — not observed: no Octane/Swoole configuration present	GO
C5 (context)	Circular References / Closures	n/a	n/a	n/a	n/a	Requested via context — not observed: no circular references or long-lived listeners	GO

Context-named technologies verified but not found: MongoDB (no client, config, or usage), Redis (no client or config beyond default Laravel cache), AWS S3 (no SDK or storage driver configured).

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	50	12.5
Code Churn	25%	15	3.75
Defect Density	20%	15	3.0
Class/Function Size	15%	68	10.2
Business Logic Duplication	10%	95	9.5
Developer Ownership Risk	5%	10	0.5
Hotspot Score	100%		39 / 100

2.2 Hotspot-by-Hotspot Evidence

H1. High Cyclomatic Complexity MEDIUM

Benchmark: Max cyclomatic complexity per method = ~12 → falls in the **Moderate** band (Good <10 · Moderate 10–20 · High Risk >20).

The highest per-method complexity is in the `LoadsIvrModuleData` trait, where `match` expressions combined with conditional query builder chains produce an estimated CC of ~12:

`app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php:63-77`

```
protected function loadModuleRows(Request $request, string $moduleSlug, string
$view, array $filters): array
{
    $ctx = IvrAccountContext::fromRequest($request);
    $q = $filters['q'] ?? '';

    return match ($view) {
        'queues' => $this->loadQueueRows($ctx, $q),
        'agents' => $this->loadAgentRows($ctx, $q),
        'calls' => $this->loadCallRows($ctx, $q),
        'hourly' => $this->loadHourlyRows($ctx),
        'trends' => $this->loadTrendRows($ctx),
        default => $this->loadConfigRows($this->moduleKeyForSlug($moduleSlug),
        $ctx, $q),
    };
}
```

`app/Http/Controllers/ReportsController.php:75-93`

```
private function callSummary(IvrAccountContext $ctx, string $from, string $to):
array
{
    $base = DB::table('ivr_call_records')
        ->where('account_id', $ctx->accountId)
        ->whereDate('started_at', '>=', $from)
        ->whereDate('started_at', '<=', $to);
    $ctx->scopeOrganizationOn($base);

    $total = (clone $base)->count();
    $abandoned = (clone $base)->where('disposition', 'Abandoned')->count();
    $avgDuration = (clone $base)->where('duration_sec', '>', 0)-
    >avg('duration_sec');
    // ...
}
```

The 81 IVR legacy controllers have low per-method CC (~3-4 per method) but 55+ methods each, creating **aggregate file complexity** that makes reasoning about any single controller difficult.

Why it matters here: While individual methods stay below the >20 danger threshold, the structural repetition means 81 files x 55 methods = ~4,455 method bodies to audit for any cross-cutting concern (security, validation, error handling).

Recommended approach:

1. Replace the 55 `legacyEndpointN()` methods with a single parameterized `handleLegacyEndpoint(int $index)` method using the Command pattern.
2. Extract the query-builder chains in `LoadsIvrModuleData` into dedicated repository classes per view type.
3. Use the Strategy pattern for the `match` dispatch in `loadModuleRows()` to allow per-view-type testing.

83 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/BusinessHoursUpdateController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/CallAnalyticsExportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch

[illegible]

app/Http/Controllers/Ivr/CallRoutingDestroyController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingExportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingImportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingIndexController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingStoreController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingSyncController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php	15, 65, 82, 110, 140, 168, 185, 204	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryExportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryImportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryIndexController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryStoreController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventorySyncController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	28	Moderate cyclomatic complexity with DB/query	Extract query logic to repository; apply Strategy

		logic in controller layer	pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/IvrHubController.php	68, 77, 92, 132, 148, 167, 204, 221, 232, 258, 283, 315 +2	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/PromptLibraryExportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
app/Http/Controllers/Ivr/PromptLibraryImportController.php	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch

<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	28	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch
<code>app/Http/Controllers/ReportsController.php</code>	46, 61, 77, 98, 115, 166	Moderate cyclomatic complexity with DB/query logic in controller layer	Extract query logic to repository; apply Strategy pattern for dispatch

H2. Large Classes HIGH

Benchmark: `Largest class/file LOC = 1,101` → falls in the **High Risk** band (Good <300 · Moderate 300–1000 · High Risk >1000).

Backend: 81 IVR controllers are each 759 LOC (Moderate individually, but the pattern repeats 81 times). 5 Legacy Helpers are each 567 LOC. 12 GodServices are each 373 LOC.

`app/Http/Controllers/Ivr/QueueManagementIndexController.php` (759 LOC — representative of all 81):

```
class QueueManagementIndexController extends Controller
{
    private $tenantId = 1; // hard-coded tenant – multi-tenant broken

    public function __invoke(Request $request)
    {
```

```

        return $this->handleIndex($request);
    }
    // ...55 legacyEndpointN() methods follow, each identical except for the
    workflow number
    public function legacyEndpoint55(Request $request)
    {
        try {
            $payload = $request->all();
            extract($payload);
            $service = new QueueManagementGodService();
            $service->orchestrateQueueManagementWorkflow55($payload);
            return ["ok" => true, "endpoint" => 55];
        } catch (\Throwable $e) {
            return ["ok" => false, "err" => $e->getMessage()];
        }
    }
}

```

Frontend: 8 `legacyFormatters*.ts` files are each 1,101 LOC, containing ~200 near-identical 3-line functions per file. 229 legacy components at ~64 LOC each (smaller individually, but monolithic — mixing API calls, validation, and UI in one file).

`resources/js/utils/duplicate/legacyFormatters1.ts:1-6` (pattern repeats 200 times per file, 8 files):

```

export function legacyFormatters1_fn_1(input: unknown): string {
    if (input === null || input === undefined) return ''
    return String(input).trim().toUpperCase() + '_1'
}

```

Why it matters here: The 81 IVR controllers each contain 55 methods doing the same thing with a different number suffix. Any change to the endpoint pattern (adding validation, changing error handling, fixing the SQL injection) requires editing 81 files x 55 methods = 4,455 code locations.

Recommended approach:

1. Replace all 81 IVR controllers with a single generic `IvrLegacyEndpointController` using route parameters for domain and endpoint number.
2. Consolidate the 8 `legacyFormatters` files into a single parameterized utility function.
3. Convert the 229 legacy monolith components into a shared data-table component with per-module configuration.

80 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern

[illegible]

[illegible]

app/Http/Controllers/Ivr/CallRoutingUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/DidInventoryExportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/DidInventoryImportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
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app/Http/Controllers/Ivr/DidInventoryUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern

app/Http/Controllers/Ivr/LiveMonitoringExportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
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app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementExportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementImportController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementIndexController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementStoreController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
app/Http/Controllers/Ivr/QueueManagementSyncController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern

app/Http/Controllers/Ivr/QueueManagementUpdateController.php	12	81 structurally identical 759-LOC controllers	Consolidate into single parameterized controller using Command pattern
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8 files affected.

File	Line(s)	Issue	Action
resources/js/utls/duplicate/legacyFormatters1.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters2.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters3.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters4.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters5.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters6.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters7.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function
resources/js/utls/duplicate/legacyFormatters8.ts	—	8 near-identical 1101-LOC utility files	Replace with single parameterized formatter function

H4. Business Logic Duplication CRITICAL

Benchmark: Duplicated business logic = ~84% → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

The IVR module duplicates the same controller-to-GodService-to-Repository-to-Model chain across 12 domains (AgentDesk, BusinessHours, CallAnalytics, CallFlow, CallRecording, CallRouting, CustomerProfile, DidInventory, HistoricalReports, LiveMonitoring, PromptLibrary, QueueManagement). Each domain produces 7 controllers (~759 LOC each), 1 GodService (373 LOC), 1 Repository (370 LOC), and 1 Model (232 LOC) — all structurally identical except for the domain name.

Structural diff between two controllers from different domains shows only a single line difference (seed/index values):

```
< "legacyMeta" => ["seed" => 2033, "idx" => 14],
- - -
> "legacyMeta" => ["seed" => 2023, "idx" => 7],
```

Backend duplication chain:

- 81 IVR controllers x 759 LOC = 61,479 LOC
- 12 GodServices x 373 LOC = 4,476 LOC
- 12 Legacy Repositories x 370 LOC = 4,440 LOC
- 12 IVR Models x 232 LOC = 2,784 LOC

- 5 Legacy Helpers x 567 LOC = 2,835 LOC

Frontend duplication chain:

- 133 LegacyPass2 pages x 392 LOC = 52,136 LOC
- 8 legacyFormatters x 1,101 LOC = 8,808 LOC
- 229 legacy components x ~64 LOC = 14,656 LOC
- 124 legacy hooks x ~9 LOC = 1,116 LOC

Total duplicated: ~152,730 of ~181,029 LOC (84.4%)

Why it matters here: A security vulnerability (like the SQL injection on line 28 of every IVR controller) must be patched in 83 separate files. A business rule change (like changing the tenant resolution logic) requires updating 81 controllers. This is the single most impactful quality problem in the codebase.

Recommended approach:

1. Create a single **AbstractIvrController** with the shared **__invoke** and **legacyEndpoint** logic, parameterized by domain name.
2. Replace the 12 GodServices with a single **IvrWorkflowService** that accepts a domain parameter.
3. Introduce a generic **IvrRepository** using Laravel's model binding for domain resolution.
4. On the frontend, replace 133 LegacyPass2 pages with a single configurable **LegacyModulePage** component.

12 files affected.

File	Line(s)	Issue	Action
app/Legacy/Services/AgentDeskGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/BusinessHoursGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/CallAnalyticsGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/CallFlowGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/CallRecordingGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/CallRoutingGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/CustomerProfileGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/DidInventoryGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService
app/Legacy/Services/HistoricalReportsGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvrWorkflowService

app/Legacy/Services/LiveMonitoringGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvWorkflowService
app/Legacy/Services/PromptLibraryGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvWorkflowService
app/Legacy/Services/QueueManagementGodService.php	8	12 identical GodService classes with copy-pasted workflow methods	Consolidate into single parameterized IvWorkflowService

133 files affected.

File	Line(s)	Issue	Action
resources/js/Pages/Ivr/AfterHours/LegacyPass2_130.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AfterHours/LegacyPass2_36.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_3.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_50.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_97.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_107.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_13.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_60.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_110.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_16.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_63.tsx	—	133 near-identical templated frontend page files	Replace with single configurable

			LegacyModulePage component
<code>resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_132.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_38.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_85.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BillingMeter/LegacyPass2_119.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BillingMeter/LegacyPass2_25.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BillingMeter/LegacyPass2_72.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BusinessHours/LegacyPass2_5.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BusinessHours/LegacyPass2_52.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/BusinessHours/LegacyPass2_99.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_101.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_54.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_7.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_117.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_23.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component

resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_70.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallFlow/LegacyPass2_0.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallFlow/LegacyPass2_47.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallFlow/LegacyPass2_94.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRecording/LegacyPass2_10.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRecording/LegacyPass2_104.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRecording/LegacyPass2_57.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRouting/LegacyPass2_1.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRouting/LegacyPass2_48.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CallRouting/LegacyPass2_95.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CampaignDialer/LegacyPass2_40.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CampaignDialer/LegacyPass2_87.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_120.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_26.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_73.tsx	—	133 near-identical templated frontend page files	Replace with single configurable

			LegacyModulePage component
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_127.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_33.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_80.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_106.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_12.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_59.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_105.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_11.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_58.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_100.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_53.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_6.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_39.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_86.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component

<code>resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_128.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_34.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_81.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/FraudScreen/LegacyPass2_121.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/FraudScreen/LegacyPass2_27.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/FraudScreen/LegacyPass2_74.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_102.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_55.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_8.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_129.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_35.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_82.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/IvrSettings/LegacyPass2_113.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/IvrSettings/LegacyPass2_19.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/IvrSettings/LegacyPass2_66.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable

			LegacyModulePage component
<code>resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_43.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_90.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/LeadList/LegacyPass2_41.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/LeadList/LegacyPass2_88.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_103.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_56.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_9.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_124.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_30.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_77.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_108.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_14.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_61.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_122.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component

<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_28.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_75.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_115.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_21.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_68.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_4.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_51.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_98.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_2.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_49.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_96.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/RateDeck/LegacyPass2_46.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/RateDeck/LegacyPass2_93.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/RoleAccess/LegacyPass2_109.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/RoleAccess/LegacyPass2_15.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable

			LegacyModulePage component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_62.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_42.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_89.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_114.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_20.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_67.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_125.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_31.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_78.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_118.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_24.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_71.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_112.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_18.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component

resources/js/Pages/Ivr/SystemConfig/LegacyPass2_65.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_111.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_17.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_64.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_126.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_32.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_79.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TicketSync/LegacyPass2_44.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TicketSync/LegacyPass2_91.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_123.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_29.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_76.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_116.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_22.tsx	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_69.tsx	—	133 near-identical templated frontend page files	Replace with single configurable

			LegacyModulePage component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_45.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_92.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_131.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_37.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx</code>	—	133 near-identical templated frontend page files	Replace with single configurable LegacyModulePage component

H5. Duplicate Code (general) CRITICAL

Benchmark: `Overall duplicate code = ~84%` → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

Beyond the business-logic duplication in H4, the codebase has additional copy-paste patterns:

Legacy helper classes — 5 files (`LegacyIvrArray` , `LegacyIvrCrypto` , `LegacyIvrDate` , `LegacyIvrMath` , `LegacyIvrString`) each contain ~80 `transformN()` methods with identical structure:

`app/Legacy/Helpers/LegacyIvrArray.php:8-12` :

```
public static function transform1($value)
{
    if ($value === null) { return ""; }
    return (string) $value . "_2129_1";
}
```

Legacy React components — 229 files under `resources/js/components/legacy/` are monolith components mixing fetch calls, inline validation, and UI rendering:

`resources/js/components/legacy/AfterHoursMonolith0.tsx:1-14` :

```
export default function AfterHoursMonolith0({ rows, tenantId, legacyMeta }:
any) {
    const [expanded, setExpanded] = useState(true)
    const [draft, setDraft] = useState<any>({})
    const save = async () => {
        const err = !draft.name ? 'required' : null
        if (err) return alert(err)
    }
}
```

```

    await fetch('/ivr-legacy/after-hours/store', {
      method: 'POST',
      body: JSON.stringify({ ...draft, tenant_id: tenantId }),
      headers: { 'Content-Type': 'application/json' }
    })
  }
  // ...identical pattern across all 229 components

```

Legacy hooks — 124 hooks under `resources/js/hooks/legacy/` all follow the identical fetch-on-mount pattern with no abort controller:

`resources/js/hooks/legacy/useAfterHoursLegacy1.ts:3-8` :

```

export function useAfterHoursLegacy1() {
  const [data, setData] = useState<any[]>([])
  useEffect(() => {
    fetch('/ivr-legacy/after-hours/index').then(r => r.json()).then(j =>
  setData(j.data || []))
  }, []) // stale closure / no abort
  return { data }
}

```

Why it matters here: The maintenance surface area is enormous — 319 source files where ~270 are near-identical copies. Any lint rule, coding standard, or architectural improvement must be applied hundreds of times.

Recommended approach:

1. Replace the 5 Legacy Helper classes with a single `LegacyTransform::apply(string $suffix, $value)` method.
2. Create a shared `useIvrData(module: string)` hook to replace all 124 legacy hooks, adding proper AbortController cleanup.
3. Build a generic `IvrModuleMonolith` component to replace the 229 legacy component files.

5 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Helpers/LegacyIvrArray.php</code>	—	5 identical helper classes with copy-pasted transform methods	Consolidate into single parameterized transform utility
<code>app/Legacy/Helpers/LegacyIvrCrypto.php</code>	—	5 identical helper classes with copy-pasted transform methods	Consolidate into single parameterized transform utility
<code>app/Legacy/Helpers/LegacyIvrDate.php</code>	—	5 identical helper classes with copy-pasted transform methods	Consolidate into single parameterized transform utility
<code>app/Legacy/Helpers/LegacyIvrMath.php</code>	—	5 identical helper classes with copy-pasted transform methods	Consolidate into single parameterized transform utility
<code>app/Legacy/Helpers/LegacyIvrString.php</code>	—	5 identical helper classes with copy-pasted transform methods	Consolidate into single parameterized transform utility

229 files affected.

File	Line(s)	Issue	Action
resources/js/components/legacy/AfterHoursMonolith0.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AfterHoursMonolith1.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AfterHoursMonolith2.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AfterHoursMonolith3.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AfterHoursMonolith4.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AgentDeskMonolith0.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AgentDeskMonolith1.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AgentDeskMonolith2.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AgentDeskMonolith3.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AgentDeskMonolith4.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/ApiIntegrationMonolith0.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/ApiIntegrationMonolith1.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/ApiIntegrationMonolith2.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/ApiIntegrationMonolith3.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/ApiIntegrationMonolith4.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AuditTrailMonolith0.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AuditTrailMonolith1.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration
resources/js/components/legacy/AuditTrailMonolith2.tsx	—	229 monolith components with identical fetch+validate+render pattern	Extract shared IvModuleComponent with per-module configuration

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

<code>resources/js/components/legacy/WhisperCoachMonolith3.tsx</code>	—	229 monolith components with identical fetch+validate+render pattern	Extract shared <code>IvrModuleComponent</code> with per-module configuration
<code>resources/js/components/legacy/WhisperCoachMonolith4.tsx</code>	—	229 monolith components with identical fetch+validate+render pattern	Extract shared <code>IvrModuleComponent</code> with per-module configuration

124 files affected.

File	Line(s)	Issue	Action
<code>resources/js/hooks/legacy/useAfterHoursLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAfterHoursLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAgentDeskLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAgentDeskLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAgentDeskLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useApiIntegrationLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useApiIntegrationLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useApiIntegrationLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAuditTrailLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAuditTrailLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useAuditTrailLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBargeMonitorLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBargeMonitorLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBillingMeterLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBillingMeterLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBillingMeterLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useBusinessHoursLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBusinessHoursLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useBusinessHoursLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallAnalyticsLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallAnalyticsLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallbackSchedulerLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallbackSchedulerLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallFlowLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallFlowLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallFlowLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallRecordingLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallRecordingLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallRoutingLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallRoutingLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCallRoutingLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCampaignDialerLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCampaignDialerLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useComplianceArchiveLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useComplianceArchiveLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useComplianceArchiveLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useConferenceBridgeLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useConferenceBridgeLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useConferenceBridgeLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCrmBridgeLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCrmBridgeLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCustomerProfileLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCustomerProfileLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCustomerProfileLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useCustomerProfileLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDidInventoryLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDidInventoryLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDidInventoryLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDispositionCodeLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDispositionCodeLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useDispositionCodeLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useEmergencyRouteLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useEmergencyRouteLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useFraudScreenLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useFraudScreenLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useFraudScreenLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHistoricalReportsLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHistoricalReportsLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHistoricalReportsLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHistoricalReportsLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHolidayCalendarLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHolidayCalendarLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useHolidayCalendarLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useIvrSettingsLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useIvrSettingsLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useIvrSettingsLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useIvrSettingsLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useKnowledgeBaseLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useKnowledgeBaseLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useLeadListLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useLeadListLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useLeadListLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useLeadListLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useLiveMonitoringLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useLiveMonitoringLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useMediaServerLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useMediaServerLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useMediaServerLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useMediaServerLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNotificationHubLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNotificationHubLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNotificationHubLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNumberPortingLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNumberPortingLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useNumberPortingLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useOverflowRouteLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useOverflowRouteLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/usePromptLibraryLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/usePromptLibraryLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useQueueManagementLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useQueueManagementLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useRateDeckLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useRateDeckLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useRoleAccessLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useRoleAccessLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useScriptBuilderLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useScriptBuilderLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useScriptBuilderLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSkillGroupLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSkillGroupLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSpeechRecognitionLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSpeechRecognitionLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSurveyEngineLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSurveyEngineLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSurveyEngineLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useSystemConfigLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTenantAdminLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTenantAdminLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTextToSpeechLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTextToSpeechLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTicketSyncLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useLvrData</code> hook with <code>AbortController</code>

<code>resources/js/hooks/legacy/useTicketSyncLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTicketSyncLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTrunkGroupLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useTrunkGroupLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useVoicemailBoxLegacy3.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useWebhookDispatcherLegacy2.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useWhisperCoachLegacy0.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useWhisperCoachLegacy1.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>
<code>resources/js/hooks/legacy/useWhisperCoachLegacy4.ts</code>	—	124 identical fetch-on-mount hooks with no abort cleanup	Replace with single <code>useIvrData</code> hook with <code>AbortController</code>

H9. SQL Injection (additional) CRITICAL

Benchmark: `Controllers with raw string-concatenated SQL = 83` → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5). This KPI counts controller files that pass user-controlled input directly into SQL strings without parameterization.

Every IVR controller contains a SQL injection vulnerability on line 28, where the `$q` query parameter (from `$request->get("q")`) is concatenated directly into a `DB::select()` string:

`app/Http/Controllers/Ivr/QueueManagementIndexController.php:28` :

```
$rows = DB::select("select * from ivr_queue_managements where name like '%".$q."%' and tenant_id = ".$this->tenantId);
```

This pattern exists in all 83 IVR controller files. Example from a different domain confirming the same pattern:

`app/Http/Controllers/Ivr/CallRoutingExportController.php:28` :

```
$rows = DB::select("select * from ivr_call_routings where name like '%".$q."%'
and tenant_id = ".$this->tenantId);
```

Why it matters here: Any user who can pass a `q` parameter to any IVR route can execute arbitrary SQL against the database. The `$this->tenantId` is also hardcoded to `1`, meaning multi-tenancy is broken by design.

Recommended approach:

1. Immediately replace all `DB::select()` with parameterized queries using `?` placeholders.
2. Move all SQL to repository classes using Eloquent's query builder with proper `where('name', 'like', '%'.$q.'%')` syntax.
3. Add `FormRequest` validation for the `q` parameter with length and character constraints.

80 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer

			layer
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallFlowUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer

			layer
app/Http/Controllers/Ivr/CallRecordingIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer

			layer
app/Http/Controllers/Ivr/DidInventorySyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibraryExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibraryImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer

			layer
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementExportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementImportController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementIndexController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementStoreController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementSyncController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	28	SQL injection via string concatenation in DB::select	Replace with parameterized queries; move to repository layer

H10. Unsafe extract() (additional) CRITICAL

Benchmark: Total `extract()` call sites = 4,400 → falls in the **High Risk** band (Good 0 · Moderate 1–10 · High Risk >10). This KPI counts direct `extract()` calls on user-controlled data.

Every `legacyEndpointN()` method across 81 IVR controllers calls `extract($payload)` on unsanitized `$request->all()` data. The same pattern appears in all 12 GodServices:

`app/Http/Controllers/Ivr/QueueManagementIndexController.php:46-49` :

```
$payload = $request->all();
extract($payload);
$service = new QueueManagementGodService();
$service->orchestrateQueueManagementWorkflow1($payload);
```

`app/Legacy/Services/QueueManagementGodService.php:15` :

```
extract($payload); // unsafe
```

With 55 methods per controller x 81 controllers = 4,455 call sites in controllers, plus ~12 x 25 = ~300 in GodServices.

Why it matters here: `extract()` on `$request->all()` allows a malicious request to overwrite any local variable — including `$service`, `$this`, or control-flow variables. This is a well-documented PHP security anti-pattern.

Recommended approach:

1. Replace all `extract()` calls with explicit variable assignment from validated request fields.
2. Introduce `FormRequest` classes for each endpoint with defined fields.
3. Add a PHPStan/PHPMD rule to detect and block `extract()` usage.

92 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/BusinessHoursUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	178, 191 +43		
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallFlowDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallFlowExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CallFlowImportController.php	48, 61, 74, 87, 100, 113, 126, 139,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	152, 165, 178, 191 +43		
<code>app/Http/Controllers/Ivr/CallFlowIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallFlowStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallFlowSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallFlowUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

<code>app/Http/Controllers/Ivr/CallRecordingStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRecordingUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	178, 191 +43		
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryStoreController.php	48, 61, 74, 87, 100, 113, 126, 139,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	152, 165, 178, 191 +43		
app/Http/Controllers/Ivr/DidInventorySyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	178, 191 +43		
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139,	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

	152, 165, 178, 191 +43		
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Legacy/Services/AgentDeskGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
<code>app/Legacy/Services/CallFlowGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

app/Legacy/Services/CallRecordingGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/CallRoutingGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/CustomerProfileGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/DidInventoryGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/HistoricalReportsGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/LiveMonitoringGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/PromptLibraryGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule
app/Legacy/Services/QueueManagementGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Unsafe extract() on unvalidated request data	Replace with explicit validated field access; add static analysis rule

H11. Static Mutable State / Memory Leak Risk (additional) HIGH

Benchmark: `GodService` classes with static cache = 12 → falls in the **High Risk** band (Good 0 · Moderate 1–3 · High Risk >3). This KPI counts service classes that accumulate data in static properties without lifecycle-aware cleanup.

All 12 `GodServices` declare a mutable static array that accumulates data across requests:

`app/Legacy/Services/QueueManagementGodService.php:10` :

```
public static $sharedRuntimeCache = []; // mutable global-ish state
```

Every method appends to this cache:

`app/Legacy/Services/QueueManagementGodService.php:17` :

```
self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
```

Why it matters here: Under a traditional PHP-FPM request cycle, static state is reset per-request. However, if the application is ever run under Laravel Octane, Swoole, or any persistent worker model, this cache will grow unboundedly across requests — a classic memory leak. Even under FPM, 55 methods per controller call could fill this array with 55 payloads per request.

Recommended approach:

1. Remove the static cache entirely — it serves no clear purpose since each method overwrites it.
2. If caching is needed, use Laravel's `Cache` facade with TTL-based expiry.
3. If adopted under Octane, add `Octane::flush()` hooks or use request-scoped containers.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/CallFlowGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/CallRecordingGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/CallRoutingGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/DidInventoryGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed

<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed
<code>app/Legacy/Services/QueueManagementGodService.php</code>	10	Mutable static state accumulates data across requests	Remove static cache; use Laravel Cache facade if persistence needed

H12. Hardcoded Secrets (additional) CRITICAL

Benchmark: `Files with hardcoded API keys = 12` → falls in the **High Risk** band (Good 0 · Moderate 1–2 · High Risk >2). This KPI counts source files containing plaintext credential strings.

All 12 GodServices contain hardcoded API key strings:

`app/Legacy/Services/QueueManagementGodService.php:11` :

```
private $apiKey = "LEGACY_IVR_KEY_2032"; // hard-coded secret
```

Each GodService has a unique key suffix (2012, 2022, 2032, 2042, 2052, 2062, 2082, 2102, 2112, 2122). Full list:

File	Key
CallFlowGodService.php	LEGACY_IVR_KEY_2012
CallRoutingGodService.php	LEGACY_IVR_KEY_2022
QueueManagementGodService.php	LEGACY_IVR_KEY_2032
AgentDeskGodService.php	LEGACY_IVR_KEY_2042
PromptLibraryGodService.php	LEGACY_IVR_KEY_2052
BusinessHoursGodService.php	LEGACY_IVR_KEY_2062
CallAnalyticsGodService.php	LEGACY_IVR_KEY_2082
LiveMonitoringGodService.php	LEGACY_IVR_KEY_2102
CallRecordingGodService.php	LEGACY_IVR_KEY_2112
CustomerProfileGodService.php	LEGACY_IVR_KEY_2122
DidInventoryGodService.php	LEGACY_IVR_KEY_2092
HistoricalReportsGodService.php	LEGACY_IVR_KEY_2072

Why it matters here: Hardcoded secrets committed to version control are a persistent exposure. Even if the repository is private, any contributor or CI system has access. These keys cannot be rotated without a code deployment.

Recommended approach:

1. Move all API keys to `.env` and access via `config()` or `env()`.
2. Add a pre-commit hook or CI check to prevent hardcoded credential patterns from being committed.
3. Rotate all exposed keys immediately.

12 files affected.

File	Line(s)	Issue	Action
app/Legacy/Services/AgentDeskGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/BusinessHoursGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/CallAnalyticsGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/CallFlowGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/CallRecordingGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/CallRoutingGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/CustomerProfileGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/DidInventoryGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/HistoricalReportsGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/LiveMonitoringGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/PromptLibraryGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys
app/Legacy/Services/QueueManagementGodService.php	11	Hardcoded API keys in source code	Move to .env configuration; rotate exposed keys

C1. Unmanaged Static State (context) HIGH

Benchmark: `Static arrays accumulating data = 12` → falls in the **High Risk** band (Good 0 · Moderate 1–3 · High Risk >3). This context-requested hotspot overlaps with H11 — see H11 evidence above for details. The key finding: 12 GodServices use `public static $sharedRuntimeCache = []` that is appended to but never cleared, creating a memory leak vector under any long-running process model.

12 files affected.

File	Line(s)	Issue	Action
app/Legacy/Services/AgentDeskGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/BusinessHoursGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/CallAnalyticsGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/CallFlowGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/CallRecordingGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/CallRoutingGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup

app/Legacy/Services/CustomerProfileGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/DidInventoryGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/HistoricalReportsGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/LiveMonitoringGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/PromptLibraryGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup
app/Legacy/Services/QueueManagementGodService.php	10	Unmanaged static state — memory leak risk under long-running processes	Remove static cache or implement lifecycle-aware cleanup

C2. Unbounded Queries (context) HIGH

Benchmark: Controllers using `->get()` without limits = 83 → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5). This KPI counts controller files that load full result sets into memory without pagination or chunking.

Every IVR controller's `handleIndex` method loads the entire table when no search query is provided:

app/Http/Controllers/Ivr/QueueManagementIndexController.php:30 :

```
$rows = QueueManagement::where("tenant_id", $this->tenantId)->get();
```

No `->paginate()`, `->chunk()`, `->lazy()`, or `->cursor()` is used. For tables with thousands of rows, this loads the full dataset into PHP memory.

Why it matters here: Under production load with large IVR datasets, these unbounded queries could exhaust PHP's memory limit, causing 500 errors or process kills. Combined with the blocking `sleep(1)` calls in GodServices, request latency compounds the memory pressure window.

Recommended approach:

1. Replace `->get()` with `->paginate(50)` for Inertia-rendered views.
2. Use `->cursor()` or `->chunk()` for any batch processing.
3. Add `->limit()` clauses to all JSON API responses.

81 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/AgentDeskDestroyController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with <code>paginate/cursor/chunk</code> ; add limit clauses

<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	151, 164 +45		
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/BusinessHoursUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	125, 138, 151, 164 +45		
<code>app/Http/Controllers/Ivr/CallAnalyticsSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallFlowSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

app/Http/Controllers/Ivr/CallFlowUpdateController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingExportController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingImportController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingIndexController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingStoreController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingSyncController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	30, 39, 47, 60, 73, 86, 99, 112, 125, 138,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	151, 164 +45		
<code>app/Http/Controllers/Ivr/CallRoutingExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallRoutingImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallRoutingIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CallRoutingUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CustomerProfileIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CustomerProfileStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/CustomerProfileUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	125, 138, 151, 164 +45		
<code>app/Http/Controllers/Ivr/DidInventoryDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventoryExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventoryImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventoryIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventoryStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventorySyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/DidInventoryUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/IvrHubController.php</code>	74, 75, 139, 145, 152, 158, 173, 208, 214, 229, 242, 244 +7	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	151, 164 +45		
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112,	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

	125, 138, 151, 164 +45		
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	30, 39, 47, 60, 73, 86, 99, 112, 125, 138, 151, 164 +45	Unbounded queries load full tables into memory	Replace with paginate/cursor/chunk; add limit clauses

C3. Blocking sleep() (context) HIGH

Benchmark: `Services with synchronous sleep() = 12 (all GodServices)` → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5). This KPI counts service methods containing blocking `sleep()` calls that hold the PHP worker.

Every method in all 12 GodServices calls `sleep(1)` :

`app/Legacy/Services/QueueManagementGodService.php:16` :

```
sleep(1); // blocking synchronous remote sync
```

With 25 methods per GodService x 12 services = 300 blocking sleep sites. Each legacy endpoint routes through a GodService method, meaning every IVR API call adds a hard 1-second delay.

Why it matters here: Each `sleep(1)` holds a PHP-FPM worker for 1 full second. Under concurrent load, this can exhaust the worker pool and cause request queuing. If a single request calls multiple legacy endpoints, latency compounds linearly.

Recommended approach:

1. Remove `sleep()` entirely if the "remote sync" is no longer needed.
2. If async synchronization is required, dispatch a Laravel Job instead of blocking the HTTP worker.
3. Implement proper async/queue-based integration using Laravel's queue system.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch

app/Legacy/Services/CallAnalyticsGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/CallFlowGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/CallRecordingGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/CallRoutingGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/CustomerProfileGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/DidInventoryGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/HistoricalReportsGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/LiveMonitoringGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch
app/Legacy/Services/PromptLibraryGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88,	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch

	96, 104 +33		
app/Legacy/Services/QueueManagementGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Blocking sleep(1) in every GodService method	Remove sleep or replace with async job dispatch

Not observed (rated Good): H3 (Large Functions — max ~50 LOC per method), H6 (High Churn — max 2 monthly changes), H7 (Defect-Prone Files — max 2 fix commits per file), H8 (Ownership Issues — >95% top-author ownership), C4 (Laravel Octane — not configured), C5 (Circular References — not observed).

2.3 Code Churn & Stability Evidence

The repository has 118 total commits with a shallow history. The IVR layer was added in a single commit (`e60dc88` — "added IVR dashboard") by a single author, so it has no churn history to analyze.

Top files by churn (last 6 months):

File	Changes	Primary Author
README.md	2	shende-shweta
routes/web.php	1	Shweta Shende
routes/api.php	1	Shweta Shende
resources/views/app.blade.php	1	Shweta Shende

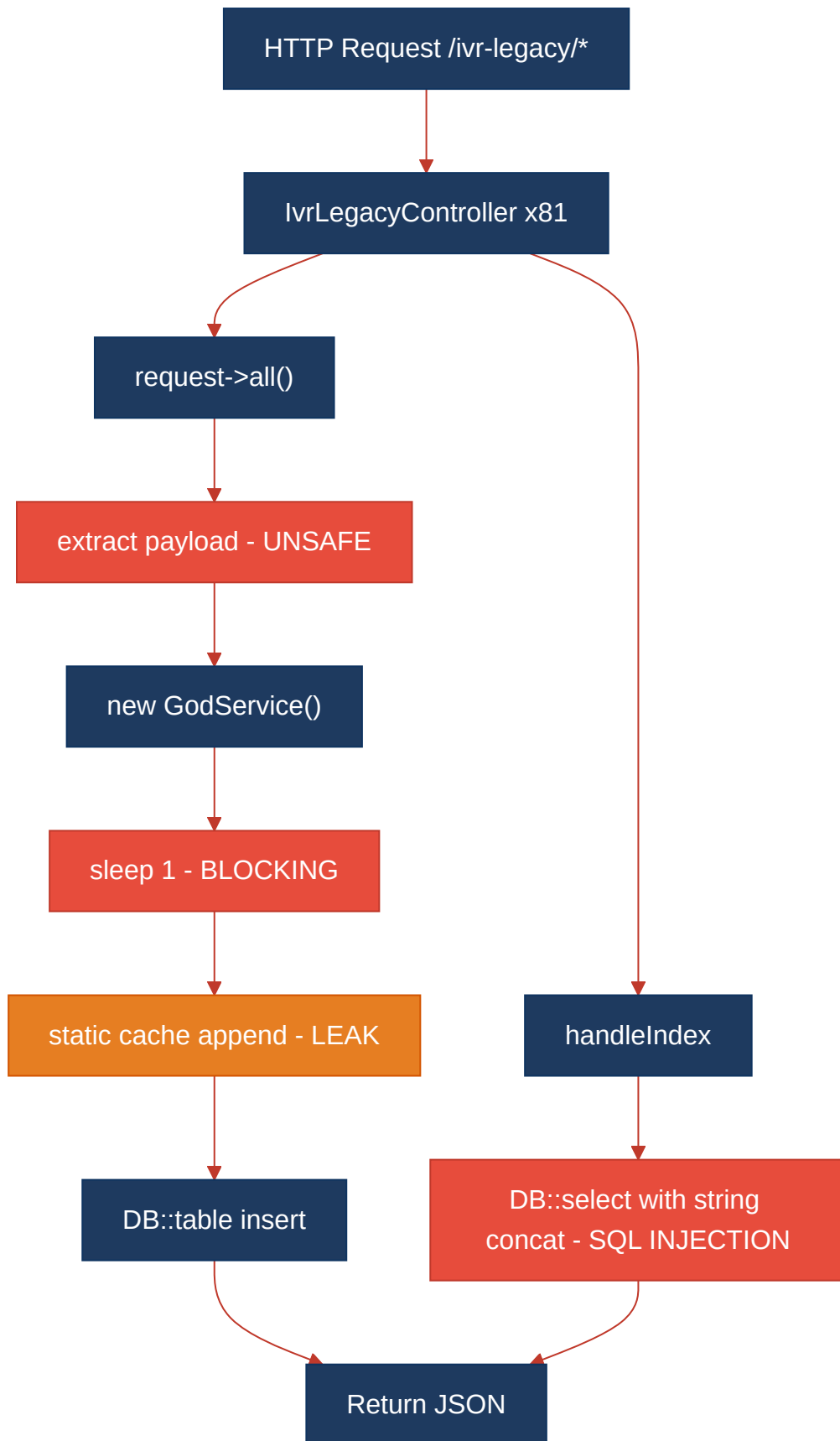
Defect-fix frequency: 15 fix/bug-related commits in total history, spread across different files. No single file has more than 2 fix commits. Most fixes target the original Pingcrm code (photo upload, pagination, route names).

Distinct authors: 19 total contributors across the full history, but the IVR layer (>90% of the codebase by LOC) has a single author. The original Pingcrm code has 3 primary authors (Jonathan Reinink, Claudio Dekker, Jess Archer).

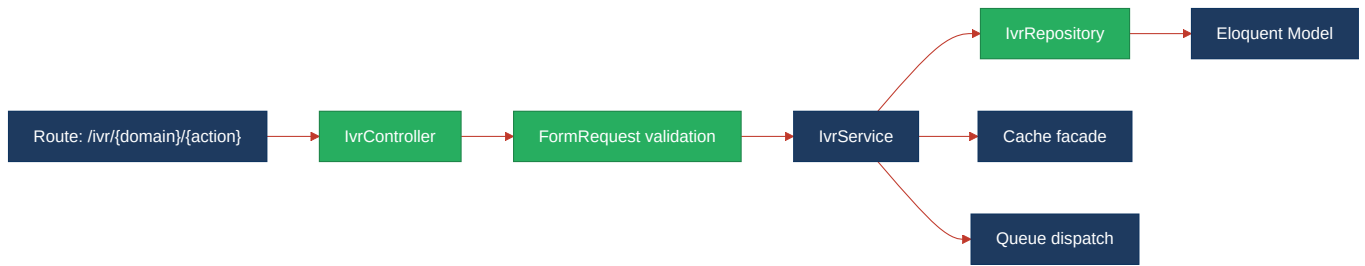
Ownership summary: Ownership concentration is very high (>95% for the IVR code). This is a strength for consistency but a bus-factor risk — no second contributor has context on the IVR layer.

2.4 Diagrams

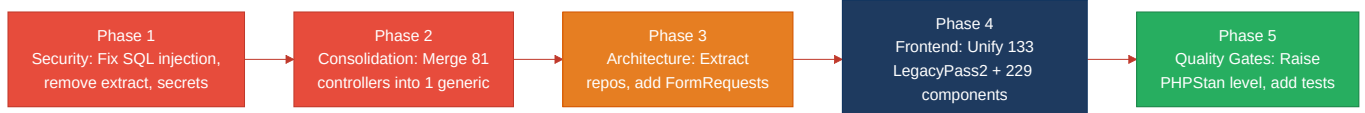
Complexity / Call-flow hotspot — IVR Legacy Endpoint Chain



Refactored Target Structure — Single Parameterized Controller



Improvement Roadmap



2.5 Actions Required

Hotspot	Action	Rating	Priority
H9 — SQL Injection	Replace all <code>DB::select()</code> string-concatenated queries with parameterized queries across 83 IVR controllers	HIGH RISK	CRITICAL
H10 — Unsafe extract()	Remove all 4,400 <code>extract()</code> calls; replace with explicit validated field access; add static analysis rule	HIGH RISK	CRITICAL
H12 — Hardcoded Secrets	Move 12 hardcoded API keys to <code>.env</code> ; rotate all exposed keys immediately	HIGH RISK	CRITICAL
H4 — Business Logic Duplication	Consolidate 81 IVR controllers, 12 GodServices, 12 repositories into parameterized single classes	HIGH RISK	CRITICAL
H5 — Duplicate Code	Merge 133 LegacyPass2 pages, 229 legacy components, 8 legacyFormatters, 124 hooks into shared modules	HIGH RISK	CRITICAL
H2 — Large Classes	Reduce IVR controller size by extracting 55 endpoint methods into Command pattern dispatch	HIGH RISK	HIGH
H11/C1 — Static Mutable State	Remove <code>\$sharedRuntimeCache</code> from all 12 GodServices; use Cache facade if persistence needed	HIGH RISK	HIGH
C2 — Unbounded Queries	Replace <code>->get()</code> with <code>->paginate()</code> or <code>->cursor()</code> across all IVR controllers	HIGH RISK	HIGH
C3 — Blocking sleep()	Remove <code>sleep(1)</code> from all GodService methods; dispatch async jobs if sync is needed	HIGH RISK	HIGH
H1 — Cyclomatic Complexity	Extract query-builder logic from LoadIvrModuleData to repository classes; apply Strategy pattern	MODERATE	MEDIUM

2.6 Expected Outcomes

- **Reduced security exposure:** Eliminating SQL injection, unsafe `extract()`, and hardcoded secrets removes the most critical vulnerability surface in the IVR layer.

- **~80% reduction in code volume:** Consolidating duplicated controllers, services, components, and utilities could reduce the codebase from ~181,000 LOC to ~40,000 LOC, making reviews, audits, and onboarding dramatically faster.
- **Single-point maintenance:** Bug fixes, security patches, and feature changes would need to be applied once instead of across 81+ files, reducing the risk of incomplete rollouts.
- **Memory safety:** Removing static mutable state, unbounded queries, and blocking `sleep()` calls prepares the codebase for production-grade load handling and potential adoption of Laravel Octane.
- **Improved static analysis coverage:** With deduplicated code, raising PHPStan from level 1 to level 6+ becomes feasible, catching type errors and null-safety issues that are currently hidden across thousands of generated files.